

Name **Suggested Solutions**



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Class	Reg Number
17	

Calculator Model / No.

Chemistry

8873/01

Paper 1

21 September 2018
1 hour

Additional Material: *Data Booklet*
OMS

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces provided at the top of this page.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the use of OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet.

Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **OMR sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

This document consists of **13** printed pages (including this cover page)

Qn	Ans
1	B
2	C
3	C
4	D
5	A
6	D
7	A
8	C
9	C
10	B

Qn	Ans
11	C
12	A
13	D
14	B
15	D
16	D
17	B
18	A
19	A
20	C

Qn	Ans
21	B
22	A
23	B
24	C
25	C
26	A
27	B
28	D
29	C
30	B

1 Answer: B

Explanation:

$$[(\text{CH}_3\text{CO}_2)_2\text{Zn}] = \frac{10.64}{183.4} = 0.0580 \text{ mol dm}^{-3}$$

$$\text{Amount of } (\text{CH}_3\text{CO}_2)_2\text{Zn} = \frac{5}{1000} \times 0.0580 = 2.90 \times 10^{-4} \text{ mol}$$



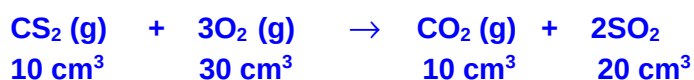
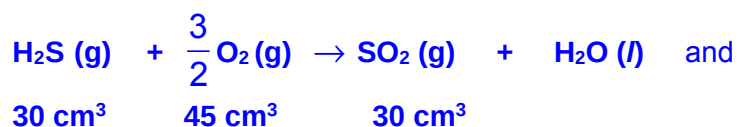
$$\text{Total amount of ions} = 3 \times 2.90 \times 10^{-4} = 8.70 \times 10^{-4} \text{ mol}$$

$$\text{Total number of ions} = 8.70 \times 10^{-4} \times 6.02 \times 10^{23} = \underline{5.2 \times 10^{20} \text{ ions}}$$

2 Answer: C

Volume of $\text{H}_2\text{S} = \frac{3}{4} \times 40 = \underline{30 \text{ cm}^3}$ and volume of $\text{CS}_2 = \frac{1}{4} \times 40 = \underline{10 \text{ cm}^3}$

Combustion equations:



Total volume of SO_2 formed = $30 + 20 = \underline{50 \text{ cm}^3}$

Volume of CO_2 formed = $\underline{10 \text{ cm}^3}$

volume of O_2 remaining = $100 - (45 + 30) = \underline{25 \text{ cm}^3}$

Volume of gaseous mixture after burning = $50 + 10 + 25 = \underline{85 \text{ cm}^3}$

Since CO_2 and SO_2 are acidic gases, they will be removed by NaOH (aq)

therefore, Volume of gaseous mixture after adding NaOH (aq) = $85 - 60 = \underline{25 \text{ cm}^3}$

3 Answer: C

By conservation of mass,

4 Answer: D

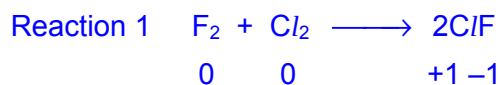
Option 1 Bonding in barium carbonate: ionic between Ba^{2+} and CO_3^{2-} , covalent within CO_3^{2-}

Option 2 Bonding in graphene: covalent only between C atoms

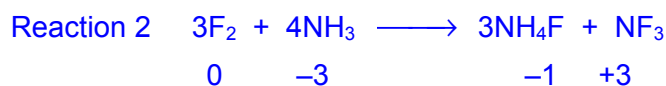
Option 3 Bonding in water: covalent only between H and O atoms

5 Answer: A

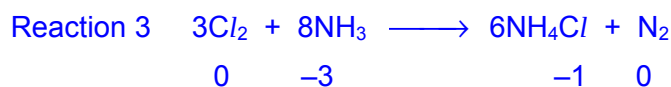
By considering change in oxidation number of element:



Cl_2 acts as reducing agent hence it is a stronger reducing agent than F_2 .



NH_3 acts as reducing agent hence it is a stronger reducing agent than F_2 .



NH_3 acts as reducing agent hence it is a stronger reducing agent than Cl_2 .

NH_3 is the strongest reducing agent whereas F_2 is the weakest reducing agent.

6 Answer: D

Statement 1 is incorrect as electrical conductivity in metals is due to the delocalised electrons, not mobile ions.

Statement 2 is correct as melting involves overcoming of the strong metallic bonds arising from the strong electrostatic forces attraction between the delocalised electrons and the residual metallic cations.

Statement 3 is correct as Al contribute more delocalised electrons than Na and this contributes to stronger metallic bonds.

7 **Answer: A**

J must be an **anion** (deflected towards positive terminal) while **L** is a **cation** (deflected towards negative plate). Options **C** and **D** are incorrect.

$$\frac{q}{m} \text{ ratio for proton } (^1\text{H}^+ \text{ nuclei}) = \frac{1}{1} \rightarrow 7.0^\circ \text{ (deflected towards negative plate)}$$

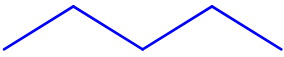
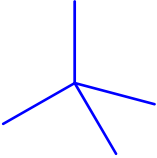
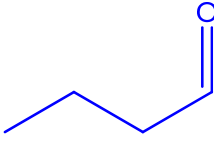
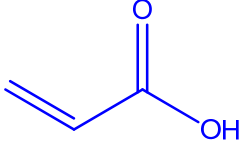
$$\frac{q}{m} \text{ ratio for } ^{14}\text{N}^- = -\frac{1}{14} \text{ hence angle of deflection of } ^{14}\text{N}^- = -\frac{1}{14} \times 7.0^\circ = -0.5^\circ \text{ (deflected towards positive plate)}$$

$$\frac{q}{m} \text{ ratio for } ^7\text{Li}^+ = +\frac{1}{7} \text{ hence angle of deflection of } ^7\text{Li}^+ = +\frac{1}{7} \times 7.0^\circ = +1.0^\circ \text{ (deflected towards negative plate)}$$

$$\frac{q}{m} \text{ ratio for } ^{18}\text{O}^- = -\frac{1}{18} \text{ hence angle of deflection of } ^{18}\text{O}^- = -\frac{1}{18} \times 7.0^\circ = -0.4^\circ \text{ (deflected towards positive plate)}$$

$$\frac{q}{m} \text{ ratio for } ^{27}\text{Al}^{3+} = +\frac{3}{27} \text{ hence angle of deflection of } ^{27}\text{Al}^{3+} = +\frac{3}{27} \times 7.0^\circ = +0.8^\circ \text{ (deflected towards negative plate)}$$

8 **Answer: C**

W	X	Y	Z
<chem>CH3CH2CH2CH2CH3</chem>	<chem>CH3C(CH3)3</chem>	<chem>CH3CH2CH2CHO</chem>	<chem>CH2=CHCOOH</chem>
 non-polar, straight chain	 non-polar, spherical	 polar (no hydrogen bonding between molecules)	 polar (hydrogen bonding present between molecules)
id-id	id-id (less extensive)	pd-pd	Hydrogen bonds
b.p. 36.1 °C	b.p. 10 °C	b.p. 74.8 °C	b.p. 141 °C

9 **Answer: C**

R is Tl and **T** is Pb. Since Tl and Pb belong to Group 13 and 14 respectively, the likely formulae of chlorides formed with Tl and Pb in this context are TlCl_3 (**RCI₃**) and PbCl_2 (**TCI₂**) respectively.

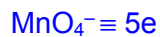
10 **Answer: B**

$$\text{Initial amount of } \text{G}^{4+} = \frac{25}{1000} \times 0.010 = 2.50 \times 10^{-4} \text{ mol}$$

Let the product formed be G^{n+}

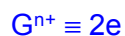
$$\text{Amount of } \text{G}^{n+} \text{ in resultant solution} = 2.50 \times 10^{-4} \text{ mol}$$

$$\text{Amount of } \text{MnO}_4^- \text{ reacted} = \frac{20}{1000} \times 0.0050 = 1.00 \times 10^{-4} \text{ mol}$$



$$\text{Amount of electrons lost} = 5.00 \times 10^{-4} \text{ mol}$$

$2.50 \times 10^{-4} \text{ mol}$ of G^{n+} gained $5.00 \times 10^{-4} \text{ mol}$ of electrons.



Hence the oxidation state of G in G^{n+} increases by 2 to +4 in G^{4+}

Therefore, G^n is G^{2+} .

11 **Answer: C**

$$\text{Half-life, } t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{\ln 2}{0.09} = 7.7 \text{ h}$$

$$\frac{C_t}{C_o} = \left(\frac{1}{2}\right)^n$$

$$\frac{18}{100} = \left(\frac{1}{2}\right)^n \Rightarrow n = 2.47$$

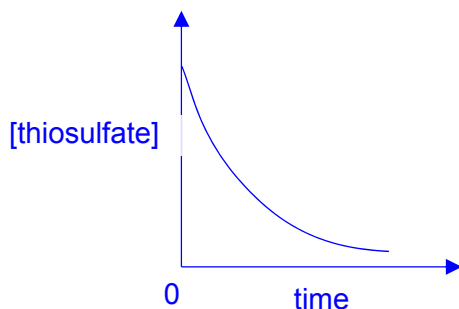
$$\text{Time taken} = \text{half-life} \times n = 7.7 \times n = \underline{\underline{19.0 \text{ h}}}$$

12 **Answer: A**

Since order of reaction w.r.t. thiosulfate is 1 $\Rightarrow$ Rate = k' [thiosulfate], where $k' = k[\text{acid}]^n$

Graph **A** shows the relationship where [thiosulfate] $\propto$ rate, while graph **B** shows a second order relationship.

Graph **C** and **D** are both incorrect as the graph of concentration against time should be

13 **Answer: D**

Bond energy of X–Y bond is the average energy absorbed to break one mole of covalent bond in the gas phase into constituent gaseous atoms under standard conditions.



Option **A** is incorrect because one mole of HF is broken into gaseous atoms and not formed from its gaseous ions.

Option **B** is incorrect because energy is not released, but absorbed when one mole of HF is broken into its gaseous atoms.

Option **C** is incorrect because the quantity in the definition of bond energy is stated as one mole of covalent bond and not one molecule.

Hence, only Option **D** best defines the term *bond energy* for a diatomic gaseous molecule.

14 **Answer: B**

$$\begin{aligned} \Delta H^\circ_{\text{rxn}} &= \sum \Delta H^\circ_f (\text{products}) - \sum \Delta H^\circ_f (\text{reactants}) \\ &= 3 \Delta H^\circ_f (\text{Fe (s)}) + \Delta H^\circ_f (\text{Al}_2\text{O}_3 (\text{s})) - 3 \Delta H^\circ_f (\text{FeO (s)}) - 2 \Delta H^\circ_f (\text{Al (s)}) \\ &= (-1676) - 3(-266) = \underline{\underline{-878 \text{ kJ mol}^{-1}}} \end{aligned}$$

15 **Answer: D**

Enthalpy change of combustion / neutralisation are always exothermic as heat is released with these reactions.

16 **Answer: D****Explanation:**

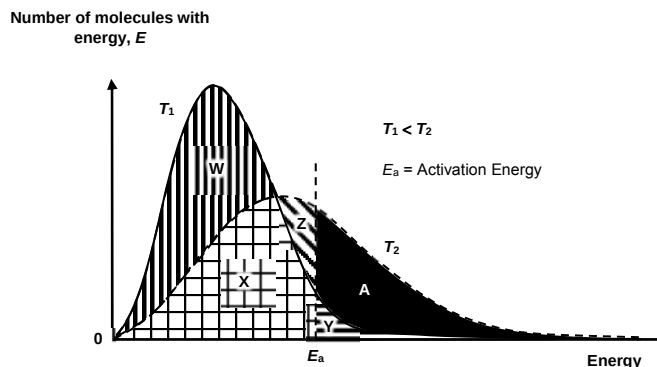
The black area (A) gives the **increase** in number of molecules with energy $\geq E_a$

The *total* number of molecules is given by the area under each graph:

$$\begin{aligned} \text{Total area under the graph} \\ = W + X + Y = X + Y + Z + A \end{aligned}$$

$$\text{So, } W = Z + A \text{ or } A = W - Z$$

$$\begin{aligned} \text{Fraction of molecules} &= \frac{A}{W + X + Y} \\ &= \frac{W - Z}{W + X + Y} \end{aligned}$$

17 **Answer: B**Option 1 is correct.

A catalyst provides an alternative reaction pathway with lower activation energy. Hence, a catalyst increases the both the rate of the forward and reverse reaction.

Option 2 is correct.

A catalyst provides an alternative reaction pathway with lower activation energy. Hence, a catalyst increases the rate of reaction through affecting the rate constant, k . (Arrhenius Equation).

Option 3 is wrong.

A catalyst does not alter the ΔH of the reaction.

Option 4 is wrong.

A catalyst does not increase the average kinetic energy of the reactant particles. Temperature does.

18 Answer: A

From the graph, the % gaseous reactants present at equilibrium decreases with an increase in pressure and temperature.

→ % product present at equilibrium increases with an increase in pressure and temperature.

This shows that equilibrium position shifts right with an increase in temperature and pressure. This implies that forward reaction is endothermic and there are more moles of gaseous reactants than products.

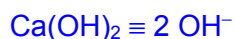
19 Answer: A

Toxic CO gas is produced in A.

20 Answer: C

$$\text{pH} = 11.5 \rightarrow \text{pOH} = 14 - 11.5 = 2.5$$

$$[\text{OH}^-] = 10^{-2.5} = 3.16 \times 10^{-3} \text{ mol dm}^{-3}$$



$$[\text{Ca(OH)}_2]_{\text{diluted}} = \frac{1}{2} \times 3.16 \times 10^{-3} = 1.08 \times 10^{-3} \text{ mol dm}^{-3}$$

Let x be the final volume of the diluted solution in cm^3

$$\text{Apply } c_1V_1 = c_2V_2 \rightarrow (0.0100) \left(\frac{10}{1000} \right) = (1.08 \times 10^{-3}) \left(\frac{x}{1000} \right)$$

$$x = 63.24 \text{ cm}^3$$

$$\text{Volume of deionised water required} = 63.24 - 10.0 = 53.2 \text{ cm}^3$$

21 Answer: B

Bromophenol-blue appears purple ($\text{pH} > 4.6$) $\Rightarrow$ Option A is **incorrect**

Phenol-red appears yellow ($\text{pH} < 6.8$) $\Rightarrow$ Options C and D are **incorrect**.

The solution must be weakly acidic and with $4.6 < \text{pH} < 6.8$.

22 Answer: A

- Atomic radius decreases from P to S to Cl.
- Melting point of S_8 is the highest among the 3 elements.
- First ionisation energy increases across the period. However 1st I.E of Mg > 1st I.E of Al as 3p valence electron of Al is further away from nucleus than 3s valence electron of Mg.
- Electrical conductivity increases with more valence electrons available for metals.

23 **Answer: B**

1	The oxide of E is amphoteric.	E is aluminum.
2	The oxide of F dissolves in water to form a strongly alkaline solution.	F is sodium.
3	The oxide of G reacts with dilute sodium hydroxide at room temperature.	G could either be phosphorus or sulfur.
4	The oxide of H is insoluble in water but is soluble in concentrated sodium hydroxide.	H is silicon.

In increasing proton number,
sodium, aluminium, silicon, phosphorus/sulfur

Option **B**: **F, E, H, G**

24 **Answer: C**

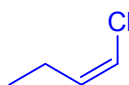
Thermal decomposition: $\text{H-X} \longrightarrow \text{H}_2 + \text{X}_2$

A is incorrect as polarity of the H–X bond does not affect thermal stability.

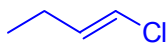
B is incorrect as the size of the H–X molecules relates to the extent of electron cloud distortion and intermolecular forces of attraction; this does not affect the H – X bond strength.

C is correct as thermal decomposition of H–X is more predominantly affected by energy required to break the H – X bond in the reactant compared with the energy released in forming the X–X bond in the product.

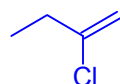
D is incorrect as thermal decomposition involves breaking the H–X bond, not the intermolecular forces of attraction between the H–X molecules.

25 Answer: C

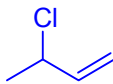
cis-1-chlorobut-1-ene



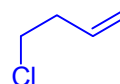
trans-1-chlorobut-1-ene



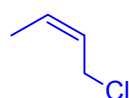
2-chlorobut-1-ene



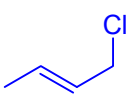
3-chlorobut-1-ene



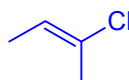
4-chlorobut-1-ene



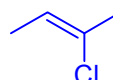
cis-1-chlorobut-2-ene



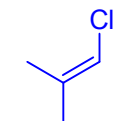
trans-1-chlorobut-2-ene



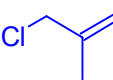
cis-2-chlorobut-2-ene



trans-2-chlorobut-2-ene



1-chloro-2-methylpropene



3-chloro-2-methylpropene

26 Answer: A

- 1 The alkene functional group in ferulic acid undergoes addition with aqueous bromine.
- 2 It is insoluble in water due to the hydrophobic benzene ring.
It undergoes neutralisation with NaOH(aq) to form sodium carboxylate salt (ionic compound) which is then soluble in water due to the formation of more favourable ion-dipole interactions.
- 3 Primary alcohol in ferulic acid undergoes oxidation with hot acidified $K_2Cr_2O_7$ to form carboxylic acid.

27 **Answer: B**

Lithium aluminium hydride will reduce both ketone and carboxylic acid to alcohols, which is **not** what happened in stage I (reduction of ketone to form secondary alcohol)

28 **Answer: D**

Ester and amide undergo basic hydrolysis simultaneously to form CH_3COONa and HOCH_2NH_2 .

29 **Answer: C**

The repeat unit of nylon-6,6 is $-\text{CO}(\text{CH}_2)_4\text{CONH}(\text{CH}_2)_6\text{NH}-$ and the 3 amide linkages undergo hydrolysis to form two monomers $\text{HO}_2\text{C}(\text{CH}_2)_4\text{CO}_2\text{H}$ and $\text{NH}_2(\text{CH}_2)_6\text{NH}_2$.

30 **Answer: B**

Statement 1 is correct. (infer from the given products).

Statement 2 is correct.

HDPE is a linear chain polymer and has little branching as the polymer chains lie close to each other and are packed regularly. The better packing in HDPE means a decrease in distance between HDPE polymer chains and hence a higher density is observed. On the other hand, LDPE has a lot of branching and this increases the distance between polymer chain molecules due to irregular packing.

Statement 3 is incorrect as the monomer is ethene not propene.

END OF PAPER